THRIVING SCHOLARS

TMUA EXAMINER-STYLE CHALLENGING MOCK

Paper 1 + Paper 2

- 40 original five-option questions
- 75 minutes per paper
- 2024–2025-informed difficulty and reasoning emphasis
- Complete concise worked solutions

SOLUTION BOOK

THRIVING SCHOLARS

Full mock, answer keys and worked solutions

Paper 1 answer key

1 E	2 A	3 B	4 A	5 E	6 B	7 C	8 A	9 E	10 C
11 B	12 E	13 D	14 C	15 A	16 D	17 E	18 D	19 B	20 C

Paper 2 answer key

1 C	2 B	3 B	4 C	5 A	6 E	7 A	8 A	9 D	10 E
11 C	12 D	13 A	14 B	15 C	16 B	17 E	18 D	19 A	20 B

PAPER 1

Applications of Mathematical Knowledge

Unfamiliar packaging, parameter reasoning, mixed-topic methods and deliberately non-monotonic difficulty.

Paper 1 · Q1 Functional equations & integration Difficulty 4/5 **Answer E**

Question 1

An integrable function f satisfies

$$f(x+1) = 2 - f(x)$$

for every real x, and

$$\int_0^1 f(x)\,dx = 3.$$

Find

$$\int_0^4 x f(x)\,dx.$$

A 8	**B** 0
C 2	**D** -8
E 4	

WORKED SOLUTION

Write

$$A = \int_0^1 f(t)\,dt = 3, \qquad B = \int_0^1 t f(t)\,dt.$$

On the intervals beginning at $0, 1, 2, 3$, the four contributions are respectively

$$B, \qquad -B, \qquad B+6, \qquad -B-2.$$

The unknown moment B cancels, giving

$$B - B + B + 6 - B - 2 = 4.$$

Answer: E.

Paper 1 · Q2

Polynomial remainders

Difficulty 4/5

Answer **A**

Question 2

When $x^{100}+1$ is divided by

$$x^2+5x+4,$$

what is the remainder?

A $\dfrac{1-4^{100}}{3}x+\dfrac{7-4^{100}}{3}$

B $\dfrac{1-4^{100}}{3}x+\dfrac{4^{100}+5}{3}$

C $\dfrac{4^{100}-1}{3}x+\dfrac{7-4^{100}}{3}$

D 2

E $\dfrac{4^{100}-1}{3}x+\dfrac{4^{100}+5}{3}$

WORKED SOLUTION

TMUA shortcut: Do not perform long division. A linear remainder is determined by its values at the two roots of the divisor.

Since

$$x^2+5x+4=(x+1)(x+4),$$

write the remainder as $R(x)=ax+b$. At the roots of the divisor, the original polynomial and the remainder have the same value:

$$R(-1)=2, \qquad R(-4)=4^{100}+1.$$

Hence

$$-a+b=2, \qquad -4a+b=4^{100}+1.$$

Subtracting gives

$$a=\frac{1-4^{100}}{3}, \qquad b=2+a=\frac{7-4^{100}}{3}.$$

Therefore the remainder is

$$\frac{1-4^{100}}{3}x+\frac{7-4^{100}}{3}.$$

Answer: A.

Paper 1 · Q3

Cubics, parameters & area

Difficulty 5/5

Answer **B**

Question 3

For $k > -4$, the curves

$$y = x^3 - 4x \qquad \text{and} \qquad y = kx$$

enclose two bounded regions. When k is increased by 2, the total enclosed area doubles. Find k.

- **A** $2\sqrt{2} - 4$
- **B** $2\sqrt{2} - 2$
- **C** $\sqrt{2} - 2$
- **D** $2 - \sqrt{2}$
- **E** $2\sqrt{2}$

WORKED SOLUTION

The intersections are $x = 0$ and $x = \pm\sqrt{k+4}$. By symmetry, the total area is

$$A(k) = 2\int_0^{\sqrt{k+4}} \left((k+4)x - x^3\right) dx = \frac{(k+4)^2}{2}.$$

The condition is

$$\frac{(k+6)^2}{2} = 2 \cdot \frac{(k+4)^2}{2}.$$

Put $t = k + 4 > 0$. Then $(t+2)^2 = 2t^2$, so $t = 2 + 2\sqrt{2}$. Therefore

$$k = t - 4 = 2\sqrt{2} - 2.$$

Answer: B.

Paper 1 · Q4

Algebraic optimisation

Difficulty 3/5

Answer **A**

Question 4

Positive real numbers a and b satisfy $a+b=12$. What is the maximum possible value of

$$ab(a-b)^2?$$

A 1296

B 1728

C 2304

D 864

E 1152

WORKED SOLUTION

Let $p=ab$. Since

$$(a-b)^2=(a+b)^2-4ab=144-4p,$$

the expression becomes

$$p(144-4p)=1296-4(p-18)^2.$$

The value $p=18$ is feasible because two positive numbers can have sum 12 and product 18. Hence the maximum is

$$1296.$$

Answer: A.

Paper 1 · Q5 Geometry & reflection | Difficulty 4/5 | Answer **E**

Question 5

The points $A=(0,0)$ and $B=(6,0)$ are fixed. A point P lies on the line

$$y=x+2.$$

What is the minimum possible value of $PA+PB$?

A 9

B $3\sqrt{10}$

C $2\sqrt{15}$

D 8

E $2\sqrt{17}$

WORKED SOLUTION

TMUA shortcut: For a shortest broken path through a line, reflect one endpoint in the line and replace the broken path by a straight segment.

Reflect $A = (0, 0)$ in $y = x + 2$. Its reflection is

$$A' = (-2, 2).$$

For every point P on the line, $PA = PA'$. Therefore

$$PA + PB = PA' + PB \geq A'B.$$

Equality is attained where the segment $A'B$ meets the line $y = x + 2$. Now

$$A'B = \sqrt{(6+2)^2 + (0-2)^2} = \sqrt{68} = 2\sqrt{17}.$$

Answer: E.

Paper 1 · Q6

Periodic sequences

Difficulty 3/5

Answer **B**

Question 6

A sequence has period 4. Its first four terms are $1, 2, 3, 4$ in an unknown order, and $a_{n+4} = a_n$. Given that

$$\sum_{n=1}^{20} n a_n = 540,$$

find $a_1 a_4 - a_2 a_3$.

- **A** -8
- **B** 2
- **C** 8
- **D** 0
- **E** -2

WORKED SOLUTION

Let

$$W = a_1 + 2a_2 + 3a_3 + 4a_4.$$

Each four-term block has sum $1 + 2 + 3 + 4 = 10$. In block j, where $j = 0, 1, 2, 3, 4$, every index is $4j$ larger than in the first block. Hence

$$540 = \sum_{j=0}^{4} (W + 4j \cdot 10) = 5W + 40(0 + 1 + 2 + 3 + 4) = 5W + 400.$$

Therefore $W = 28$. Since $a_1 + a_2 + a_3 + a_4 = 10$, subtracting this from W gives

$$a_2 + 2a_3 + 3a_4 = 18,$$

or, equivalently after using $a_4 = 10 - a_1 - a_2 - a_3$,

$$3a_1 + 2a_2 + a_3 = 12.$$

Now test the four possible values of a_1:

a_1	$2a_2 + a_3 = 12 - 3a_1$	possible?
1	9	no choice from $2, 3, 4$
2	6	$(a_2, a_3) = (1, 4)$
3	3	impossible
4	0	impossible

Thus $(a_1, a_2, a_3, a_4) = (2, 1, 4, 3)$, and

$$a_1a_4 - a_2a_3 = 2 \cdot 3 - 1 \cdot 4 = 2.$$

Answer: B.

Paper 1 · Q7 Infinite series Difficulty 3/5 **Answer C**

Question 7

Evaluate the infinite series

$$\frac{1}{2} + \frac{2}{2^2} + \frac{1}{2^3} + \frac{4}{2^4} + \frac{1}{2^5} + \frac{6}{2^6} + \cdots,$$

where every odd-position numerator is 1, while the numerator in position $2m$ is $2m$.

- **A** 2
- **B** $\frac{22}{9}$
- **C** $\frac{14}{9}$
- **D** $\frac{10}{9}$
- **E** $\frac{16}{9}$

WORKED SOLUTION

The odd-position terms form a geometric series:

$$\frac{1}{2} + \frac{1}{2^3} + \frac{1}{2^5} + \cdots = \frac{1/2}{1 - 1/4} = \frac{2}{3}.$$

For the even-position terms, put

$$T = \frac{1}{4} + \frac{2}{4^2} + \frac{3}{4^3} + \cdots.$$

Then

$$\frac{1}{4}T = \frac{1}{4^2} + \frac{2}{4^3} + \frac{3}{4^4} + \cdots.$$

Subtracting the second series from the first gives

$$\frac{3}{4}T = \frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \cdots = \frac{1/4}{1-1/4} = \frac{1}{3},$$

so $T = \frac{4}{9}$. The even-position numerators are $2m$, so their total contribution is

$$2T = \frac{8}{9}.$$

Therefore the required sum is

$$\frac{2}{3} + \frac{8}{9} = \frac{14}{9}.$$

Answer: C.

Paper 1 · Q8 Circle geometry & coordinates Difficulty 5/5 **Answer A**

Question 8

Three circles with radii 2, 3 and 10 are pairwise externally tangent. Their three points of pairwise tangency are A, B and C. What is the area of triangle ABC?

- **A** $\frac{60}{13}$
- **B** 6
- **C** 30
- **D** $\frac{30}{13}$
- **E** 5

WORKED SOLUTION

TMUA shortcut: The distances between the circle centres are $5, 12, 13$, so the centre triangle is immediately a right triangle.

Place the centres of the circles of radii $2, 3, 10$ at

$$O_1 = (0,0), \qquad O_2 = (5,0), \qquad O_3 = (0,12).$$

The tangency points involving the circle of radius 2 are

$$A = (2,0), \qquad B = (0,2).$$

The tangency point of the circles of radii 3 and 10 divides O_2O_3 in the ratio $3:10$, so

$$C = O_2 + \frac{3}{13}(O_3 - O_2) = \left(\frac{50}{13}, \frac{36}{13}\right).$$

Using A as the origin for the two side vectors,

$$[ABC] = \frac{1}{2}\left|\begin{vmatrix} -2 & 2 \\ \frac{24}{13} & \frac{36}{13} \end{vmatrix}\right| = \frac{60}{13}.$$

Answer: A.

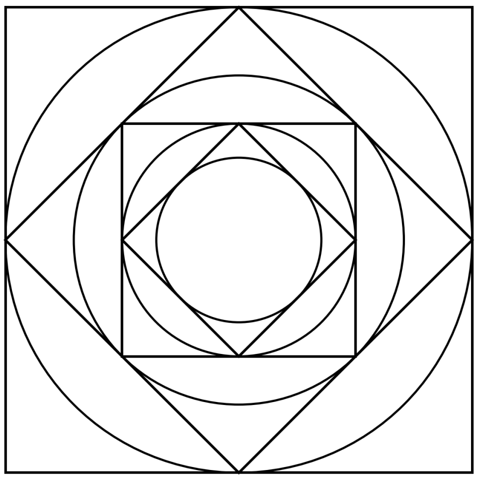
Paper 1 · Q9 Infinite geometry Difficulty 3/5 Answer **E**

Question 9

A circle is inscribed in a square of side 4. A new square is then inscribed in that circle. The process is repeated indefinitely.

At every stage, record the area inside the current square but outside its inscribed circle. What is the sum of all the recorded areas?

Paper 1, Question 9



Boundary diagram only; the regions whose areas are recorded are not shaded.

- **A** $6(4 - \pi)$
- **B** $4(4 - \pi)$
- **C** $16 - 2\pi$
- **D** $32 - 4\pi$
- **E** $8(4 - \pi)$

WORKED SOLUTION

For a square of side s, the shaded part has area

$$s^2 - \pi\left(\frac{s}{2}\right)^2 = s^2\left(1 - \frac{\pi}{4}\right).$$

Each succeeding square has half the area of the preceding square. Therefore

$$16\left(1-\frac{\pi}{4}\right)\left(1+\frac{1}{2}+\frac{1}{4}+\cdots\right)=8(4-\pi).$$

Answer: E.

Paper 1 · Q10 Trigonometric equations Difficulty 3/5 **Answer C**

Question 10

The real numbers p, q, r satisfy

$$p+q+r=0 \qquad \text{and} \qquad p>r>0.$$

How many solutions does

$$p\sin^2\theta+q\sin\theta+r=0$$

have for $0<\theta<\pi$?

- **A** 4
- **B** 2
- **C** 3
- **D** 1
- **E** 0

WORKED SOLUTION

TMUA shortcut: Use $p+q+r=0$ before applying the quadratic formula; the expression factorises.

Let $s=\sin\theta$. Since $q=-(p+r)$,

$$ps^2-(p+r)s+r=(s-1)(ps-r).$$

Hence

$$\sin\theta=1 \qquad \text{or} \qquad \sin\theta=\frac{r}{p}.$$

The first equation has one solution in $0<\theta<\pi$. Since $0<r/p<1$, the second has two solutions in that interval. The two sets do not overlap.

$$1+2=3.$$

Answer: C.

Paper 1 · Q11

Coordinate geometry | Difficulty 3/5 | Answer **B**

Question 11

A circle has centre $(p, 6)$, is tangent to the x-axis, and passes through $(4, 10)$. There are two possible values of p. What is the distance between them?

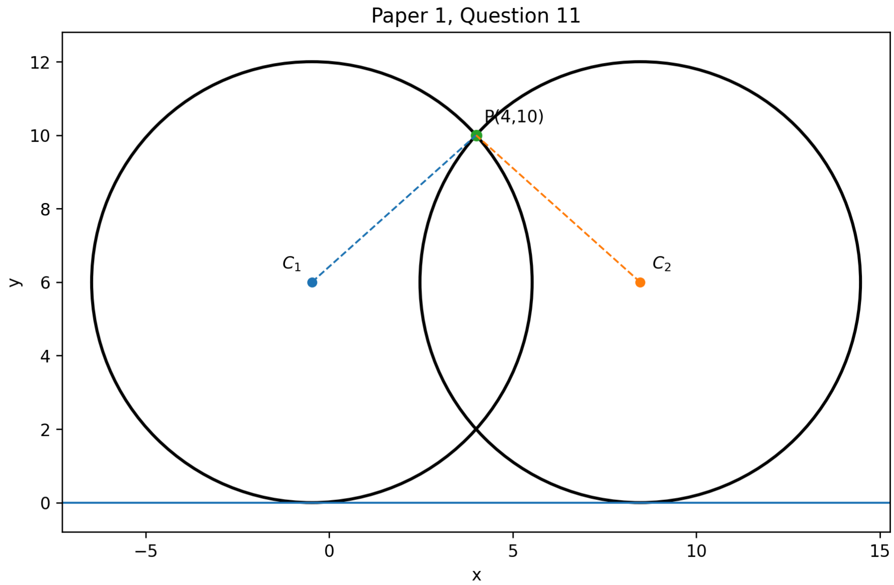


Diagram for Paper 1, Question 11 — not to scale.

A $2\sqrt{5}$

B $4\sqrt{5}$

C 10

D $6\sqrt{5}$

E 8

WORKED SOLUTION

Tangency to the x-axis gives radius 6. Hence

$$(p-4)^2 + (6-10)^2 = 6^2,$$

so $(p-4)^2 = 20$ and $p = 4 \pm 2\sqrt{5}$. Their separation is

$$4\sqrt{5}.$$

Answer: B.

Paper 1 · Q12

Function transformations & roots | Difficulty 2/5 | Answer **E**

Question 12

A function f has exactly three real zeros: $-3, 1, 5$. Define

$$g(x) = |f(2|x| - 1)|.$$

How many distinct real zeros does g have?

- **A** 5
- **B** 2
- **C** 3
- **D** 6
- **E** 4

WORKED SOLUTION

The input $2|x| - 1$ is at least -1, so the zero -3 of f cannot occur. The other two zeros give

$$2|x| - 1 = 1 \quad \text{or} \quad 2|x| - 1 = 5.$$

Thus $|x| = 1$ or $|x| = 3$, producing

$$x = \pm 1, \ \pm 3.$$

The outer modulus does not change the zeros. There are 4 distinct zeros.

Answer: E.

Paper 1 · Q13 | Counting & divisibility | Difficulty 4/5 | Answer **D**

Question 13

How many six-digit positive integers can be formed using six distinct digits selected from

$$0, 1, 2, 3, 4, 5, 6$$

and are divisible by 15?

- **A** 480
- **B** 432
- **C** 528
- **D** 552
- **E** 456

WORKED SOLUTION

The selected digit sum must be divisible by 3, and the last digit must be 0 or 5. The sum of all seven digits is 21, so the omitted digit must be $0, 3$, or 6.

- Omit 0: the final digit must be 5, giving $5! = 120$.

- Omit 3: ending in 0 gives $5! = 120$; ending in 5 gives $5! - 4! = 96$.
- Omit 6: the same count gives $120 + 96 = 216$.

$$120 + 216 + 216 = 552.$$

Answer: D.

Paper 1 · Q14 Coordinate geometry & area Difficulty 4/5 **Answer C**

Question 14

A line $y = mx + c$ passes through $(2, 5)$ and meets the positive coordinate axes, forming a triangle of area 25 with the axes. There are two possible gradients.

What is the product of these two gradients?

- **A** 5
- **B** $\frac{5}{4}$
- **C** $\frac{25}{4}$
- **D** 25
- **E** $\frac{25}{2}$

WORKED SOLUTION

The gradient is negative. Put $m = -t$, where $t > 0$. Since the line passes through $(2, 5)$,

$$c = 5 + 2t.$$

The intercepts are c and c/t, so

$$\frac{1}{2}c\left(\frac{c}{t}\right) = 25.$$

Therefore

$$(5 + 2t)^2 = 50t, \qquad 4t^2 - 30t + 25 = 0.$$

The product of the two values of t is $25/4$. Since each gradient is $-t$, their product is also

$$\frac{25}{4}.$$

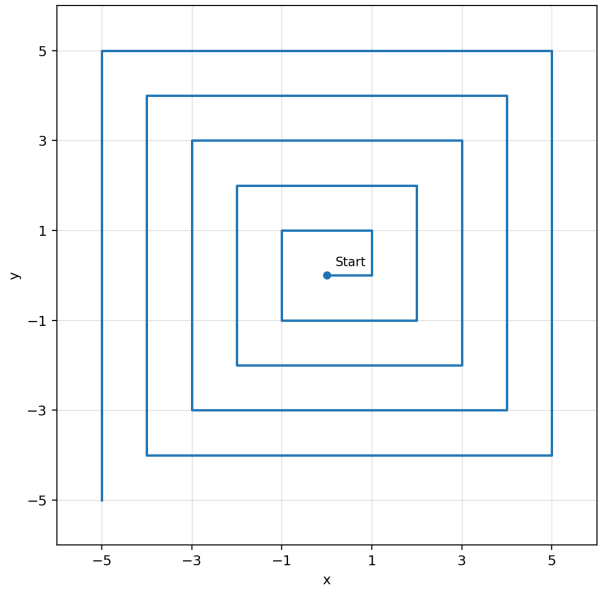
Answer: C.

Paper 1 · Q15 Coordinate patterns Difficulty 2/5 **Answer A**

Question 15

A path begins at $(0, 0)$. It moves right 1, up 1, left 2, down 2, right 3, up 3, and continues in this pattern.

Thus each positive integer length occurs on two consecutive segments, and the directions repeat right, up, left, down. What is the straight-line distance from the origin after 20 segments?



Schematic path; the final endpoint is deliberately not labelled.

A $5\sqrt{2}$

B $10\sqrt{2}$

C 5

D 10

E 20

WORKED SOLUTION

TMUA shortcut: Do not trace all 20 segments. Group the path into complete right–up–left–down blocks.

The first block uses lengths $1, 1, 2, 2$, the second uses $3, 3, 4, 4$, and so on. In every four-segment block, the horizontal displacement is

$$(2j - 1) - 2j = -1,$$

and the vertical displacement is also -1. There are $20/4 = 5$ complete blocks, so the endpoint is

$$(-5, -5).$$

Its distance from the origin is

$$\sqrt{(-5)^2 + (-5)^2} = 5\sqrt{2}.$$

Answer: A.

Paper 1 · Q16

Numerical integration

Difficulty 4/5

Answer **D**

Question 16

A quadratic function f satisfies $f(0) = 2$ and $f(4) = 18$. The ordinates at $x = 0, 1, 2, 3, 4$ are used with the trapezium rule, with strip width 1, to estimate

$$\int_0^4 f(x)\,dx.$$

The estimate exceeds the exact integral by 2. Find $f(2)$.

- **A** -6
- **B** 6
- **C** 10
- **D** -2
- **E** 2

WORKED SOLUTION

Write

$$f(x) = ax^2 + bx + 2.$$

For this quadratic, direct calculation of the composite trapezium estimate T gives

$$T - \int_0^4 f(x)\,dx = \frac{2a}{3}.$$

Hence $2a/3 = 2$, so $a = 3$. From $f(4) = 18$,

$$48 + 4b + 2 = 18, \qquad b = -8.$$

Therefore

$$f(2) = 3(4) - 8(2) + 2 = -2.$$

Answer: D.

Paper 1 · Q17

Quartics & parameter ranges

Difficulty 5/5

Answer **E**

Question 17

For which values of the real parameter k does

$$x^4 - 2(k+1)x^2 + k^2 = 0$$

have exactly four distinct real solutions?

A $k > 0$

B $k > -\frac{1}{2}$

C $-\frac{1}{2} < k < 0$

D $k \neq 0$

E $(-\frac{1}{2}, 0) \cup (0, \infty)$

WORKED SOLUTION

Put $y = x^2$. Then

$$y^2 - 2(k+1)y + k^2 = 0,$$

whose roots are

$$y = (k+1) \pm \sqrt{2k+1}.$$

Four distinct real x-solutions require two distinct positive y-roots. Distinct real roots require $k > -1/2$. Under this condition, the smaller root is positive precisely when

$$k+1 > \sqrt{2k+1} \quad \Longleftrightarrow \quad k^2 > 0.$$

Thus $k = 0$ must be excluded, giving

$$\left(-\frac{1}{2}, 0\right) \cup (0, \infty).$$

Answer: E.

Paper 1 · Q18

Radical optimisation

Difficulty 4/5

Answer D

Question 18

Find the maximum value of

$$x + \sqrt{(x+1)(9-x)}$$

for $-1 \le x \le 9$.

- **A** $4 + 3\sqrt{2}$
- **B** $9 + \sqrt{2}$
- **C** 9
- **D** $4 + 5\sqrt{2}$
- **E** $5\sqrt{2}$

WORKED SOLUTION

Let

$$F(x) = x + \sqrt{(x+1)(9-x)}.$$

At an interior stationary point,

$$F\backslash'(x) = 1 + \frac{4-x}{\sqrt{(x+1)(9-x)}} = 0.$$

Hence $x > 4$ and

$$\sqrt{(x+1)(9-x)} = x - 4.$$

Squaring gives $2x^2 - 16x + 7 = 0$, so the relevant root is

$$x = 4 + \frac{5\sqrt{2}}{2}.$$

At this point the radical equals $x - 4$, and therefore

$$F(x) = 2x - 4 = 4 + 5\sqrt{2}.$$

This exceeds the endpoint values -1 and 9.

Answer: D.

Paper 1 · Q19 Nested modulus & root counts | Difficulty 5/5 | Answer **B**

Question 19

Let $k \geq 0$. For which values of k does

$$\big| |x^2 - 4x| - 3 \big| = k$$

have exactly six distinct real solutions?

A $0 < k < 1$

B $1 < k < 3$

C $1 \leq k < 3$

D $k > 3$

E $k = 2$

WORKED SOLUTION

For $c \geq 0$, the equation $|x^2 - 4x| = c$ has

- 4 roots when $0 < c < 4$;
- 3 roots when $c = 4$;
- 2 roots when $c = 0$ or $c > 4$.

The given equation produces the levels $c = 3 + k$ and, when nonnegative, $c = 3 - k$.

For $1 < k < 3$, the first level is greater than 4, giving 2 roots, while the second lies between 0 and 2, giving 4 roots. The endpoint and remaining ranges give $8, 7, 4$, or 2 roots instead.

$$1 < k < 3.$$

Answer: B.

Paper 1 · Q20

Parametric area

Difficulty 5/5

Answer C

Question 20

The parabola $y = x^2$ and the line $y = mx + 4$ enclose an area of 36. Find m^2.

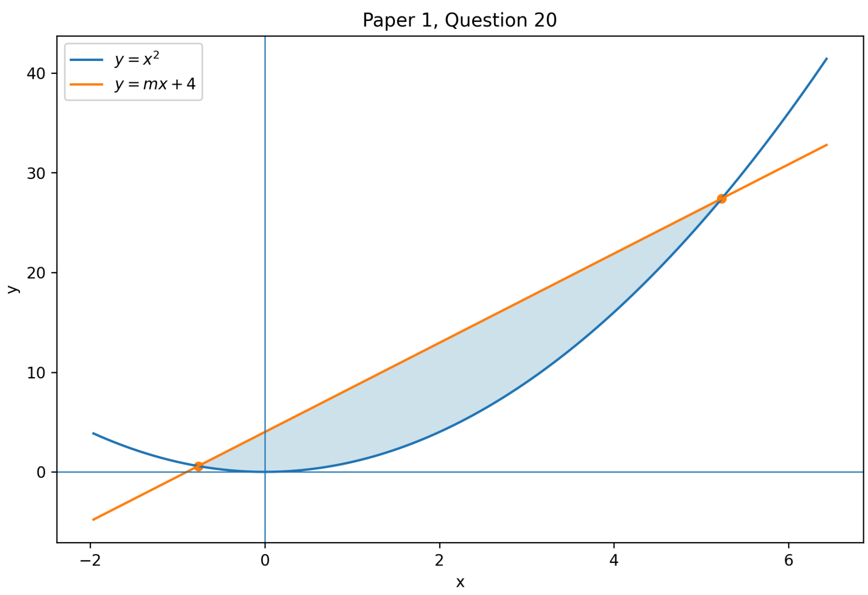


Diagram for Paper 1, Question 20 — not to scale.

A 24

B 32

C 20

D 16

E 18

WORKED SOLUTION

The intersection equation is

$$x^2 - mx - 4 = 0.$$

Let its roots be $r_1 < r_2$, with separation $d = r_2 - r_1$. Between the roots,

$$mx + 4 - x^2 = (x - r_1)(r_2 - x).$$

Therefore the enclosed area is

$$\int_0^d t(d - t)\,dt = \frac{d^3}{6}.$$

Since the area is 36, $d = 6$. The root separation is also $\sqrt{m^2 + 16}$, so

$$m^2 + 16 = 36, \qquad m^2 = 20.$$

Answer: C.

Mathematical Reasoning

Counterexamples, proof analysis, implication, quantifiers, transformed roots, divisibility and data sufficiency.

Paper 2 · Q1

Transformed roots & solution counts

Difficulty 4/5

Answer C

Question 1

A real function f has exactly four distinct real zeros

$$a < b < c < d.$$

The equation

$$f(x^2 - 4) = 0$$

has exactly five distinct real solutions. Which statement must be true?

- **A** $a < b < -4 < c < d$
- **B** $a = -4 < b < c < d$
- **C** $a < -4, \quad b = -4, \quad -4 < c < d$
- **D** $-4 < a < b < c < d$
- **E** $a < -4 < b < c < d$

WORKED SOLUTION

For a zero r of f, the corresponding equation is

$$x^2 - 4 = r, \qquad \text{so} \qquad x^2 = r + 4.$$

This contributes no real solution if $r < -4$, one solution if $r = -4$, and two solutions if $r > -4$.

Five solutions from four distinct zeros can therefore arise only as

$$0 + 1 + 2 + 2.$$

In increasing order, the zeros must satisfy

$$a < -4, \qquad b = -4, \qquad -4 < c < d.$$

Answer: C.

Paper 2 · Q2

Identifying a proof error

Difficulty 3/5

Answer **B**

Question 2

A student claims that $x^2 < y^2$ implies $x < y$, using the following argument:

Line 1. $x^2 - y^2 < 0$.

Line 2. $(x - y)(x + y) < 0$.

Line 3. Exactly one of $x - y$ and $x + y$ is negative.

Line 4. Since $x - y < x + y$, the negative factor is $x - y$.

Line 5. Hence $x - y < 0$, so $x < y$.

Which is the first incorrect line?

- **A** Line 2
- **B** Line 4
- **C** Line 1
- **D** Line 3
- **E** Line 5

WORKED SOLUTION

Lines 1–3 are valid. Line 4 assumes

$$x - y < x + y,$$

which is equivalent to $y > 0$, but no sign condition on y was given. For example, $x = 2$, $y = -3$ satisfies $x^2 < y^2$ but not $x < y$.

Answer: B.

Paper 2 · Q3

Sign changes & parity of roots

Difficulty 4/5

Answer **B**

Question 3

Let f be a real polynomial. Every real root of f is simple, and $f(0)f(1) \neq 0$. Let k be the number of real roots of f in $(0,1)$. Consider

$$P:\ f(0)f(1) < 0, \qquad Q:\ k \text{ is odd.}$$

Which describes the relationship between P and Q?

A P is necessary but not sufficient for Q

B P is necessary and sufficient for Q

C P is neither necessary nor sufficient for Q

D P is sufficient but not necessary for Q

E Q is impossible

WORKED SOLUTION

At every simple root, the sign of a polynomial changes. Moving from $x = 0$ to $x = 1$, the sign therefore changes exactly k times.

If k is odd, the endpoint signs are opposite, so $f(0)f(1) < 0$. If k is even, the endpoint signs are the same, so $f(0)f(1) > 0$.

Hence

$$f(0)f(1) < 0 \quad \Longleftrightarrow \quad k \text{ is odd.}$$

Answer: B.

Paper 2 · Q4

Triangle inequalities

Difficulty 4/5

Answer C

Question 4

Let a, b, c be positive real numbers. Consider

$$P: \ a, b, c \text{ are the sides of a nondegenerate triangle,}$$

$$Q: \ (a+b+c)^2 > 2(a^2+b^2+c^2).$$

Which is correct?

A P is necessary and sufficient for Q

B Q is true for all positive a, b, c

C P is sufficient but not necessary for Q

D P is necessary but not sufficient for Q

E P is neither necessary nor sufficient for Q

WORKED SOLUTION

If P holds, write

$$a = y + z, \qquad b = z + x, \qquad c = x + y$$

for positive x, y, z. Then

$$(a+b+c)^2 - 2(a^2+b^2+c^2) = 4(xy+yz+zx) > 0.$$

So P is sufficient. It is not necessary: $(a, b, c) = (1, 1, 2)$ does not form a nondegenerate triangle, but $16 > 12$.

Answer: C.

Paper 2 · Q5 Monotonicity & integrals Difficulty 4/5 Answer **A**

Question 5

A differentiable function f satisfies

$$f'(x) < 0 \qquad \text{for every } x \in [0,1].$$

Consider:

$$\begin{aligned} \text{I.} &\quad \int_0^{1/2} f(x)\,dx > \int_{1/2}^1 f(x)\,dx, \\ \text{II.} &\quad \int_0^1 f(x)\,dx < \int_0^1 \left(f(x)\right)^2 dx, \\ \text{III.} &\quad f(0) < f(1). \end{aligned}$$

Which statements must be true?

A I only

B I and III only

C II only

D III only

E I and II only

WORKED SOLUTION

TMUA shortcut: Compare the two half-intervals point by point by pairing x with $x + \frac{1}{2}$.

Because f is strictly decreasing, for $0 \le x \le \frac{1}{2}$,

$$f(x) > f\left(x + \frac{1}{2}\right).$$

Therefore

$$\int_0^{1/2} f(x)\,dx - \int_{1/2}^1 f(x)\,dx = \int_0^{1/2} \left[f(x) - f\left(x + \frac{1}{2}\right)\right] dx > 0.$$

So I must be true.

II need not be true. For example, $f(x) = \frac{1}{2} - \frac{x}{4}$ is strictly decreasing and satisfies $0 < f(x) < 1$, so $f(x)^2 < f(x)$ throughout the interval.

III is false for every strictly decreasing f, because $f(0) > f(1)$.

Answer: A.

Paper 2 · Q6

Nested quantifiers & negation

Difficulty 4/5

Answer **E**

Question 6

A real sequence (a_n) satisfies the statement

$$\text{For every real } M, \text{ there exists a positive integer } N \text{ such that, for every } n \geq N, \ a_n > M.$$

Which is the logical negation of this statement?

A There exists a real M such that, for every positive integer N and every $n \geq N$, $a_n \leq M$.

B For every real M, there exists a positive integer N such that, for every $n \geq N$, $a_n \leq M$.

C For every real M and every positive integer N, there exists $n \geq N$ with $a_n \leq M$.

D There exists a real M and a positive integer N such that, for every $n \geq N$, $a_n \leq M$.

E There exists a real M such that, for every positive integer N, there exists $n \geq N$ with $a_n \leq M$.

WORKED SOLUTION

Negate the quantifiers one at a time and negate the final strict inequality:

$$\neg\big(\forall M\, \exists N\, \forall n \geq N :\ a_n > M\big) \equiv \exists M\, \forall N\, \exists n \geq N :\ a_n \leq M.$$

The negation says that there is one fixed level M below which the sequence returns, however far along the sequence one begins.

Answer: E.

Paper 2 · Q7

Prime exponents & divisibility

Difficulty 3/5

Answer **A**

Question 7

Let

$$N = 2^a 3^b 5^c,$$

where a, b, c are nonnegative integers. Which condition is necessary and sufficient for 5400 to divide N^2?

A $180 \mid N$

B $360 \mid N$

C $90 \mid N$

D $120 \mid N$

E $60 \mid N$

WORKED SOLUTION

Factor

$$5400 = 2^3 3^3 5^2.$$

Since $N^2 = 2^{2a}3^{2b}5^{2c}$, divisibility requires

$$2a \geq 3, \qquad 2b \geq 3, \qquad 2c \geq 2.$$

Because the exponents are integers, this is

$$a \geq 2, \qquad b \geq 2, \qquad c \geq 1,$$

which is exactly $180 = 2^2 3^2 5 \mid N$.

Answer: A.

Paper 2 · Q8 Colouring implications Difficulty 4/5 **Answer A**

Question 8

Every integer is coloured exactly one of red or green. For every integer n:

- if n is red, then $n+2$ is green;
- if n is green, then $n+100$ is red.

Given that 205 is green, which integer must be green?

A 9

B 109

C 5

D 7

E 105

WORKED SOLUTION

Because 205 is green, it is not red. Taking the contrapositive of the second rule with $n = 105$, 105 must be red. Hence 107 is green.

Now 107 is not red. Applying the same contrapositive with $n = 7$, 7 is red, and therefore 9 is green.

Answer: A.

Paper 2 · Q9 Integral bounds & feasibility Difficulty 5/5 **Answer D**

Question 9

A function f satisfies

$$0 \le f(x) \le 1 \qquad (0 \le x \le 4),$$

together with

$$\int_0^2 f(x)\,dx = 1, \qquad \int_1^3 f(x)\,dx = \frac{3}{2}.$$

What is the range of possible values of

$$\int_0^4 f(x)\,dx?$$

A $\left(\frac{3}{2}, 3\right)$

B $[2, 3]$

C $[1, 3]$

D $\left[\frac{3}{2}, 3\right]$

E $\left[\frac{3}{2}, \frac{7}{2}\right]$

WORKED SOLUTION

TMUA shortcut: Replace the function by the four unit-interval integrals. The functional problem becomes a four-variable bounds problem.

Let

$$a = \int_0^1 f, \quad b = \int_1^2 f, \quad c = \int_2^3 f, \quad d = \int_3^4 f.$$

Since $0 \le f \le 1$, each of a, b, c, d lies in $[0, 1]$. The given information becomes

$$a + b = 1, \qquad b + c = \frac{3}{2}.$$

Thus $a = 1 - b$ and $c = \frac{3}{2} - b$. The constraints $0 \le a, b, c \le 1$ give

$$\frac{1}{2} \le b \le 1, \qquad \frac{1}{2} \le c \le 1.$$

Now

$$\int_0^4 f(x)\,dx = a + b + c + d = 1 + c + d.$$

Since $\frac{1}{2} \le c \le 1$ and $0 \le d \le 1$,

$$\frac{3}{2} \le \int_0^4 f(x)\,dx \le 3.$$

The lower endpoint is attained, for example, by the piecewise-constant function taking the values

$$0,\ 1,\ \frac{1}{2},\ 0$$

on the four successive unit intervals. The upper endpoint is attained by taking the values

$$\frac{1}{2},\ \frac{1}{2},\ 1,\ 1.$$

Both examples satisfy the two given integrals, so both endpoints are included.

Answer: D.

Paper 2 · Q10 Proof direction & inequalities Difficulty 4/5 Answer **E**

Question 10

Let a, b, c be positive real numbers. A student claims that

$$a^2 + b^2 > c^2 \quad \Longrightarrow \quad a + b > \sqrt{2}\,c,$$

using the following argument:

Line 1. $a^2 + b^2 > c^2$.
Line 2. $2ab \leq a^2 + b^2$.
Line 3. $(a+b)^2 = a^2 + b^2 + 2ab \leq 2(a^2 + b^2)$.
Line 4. Therefore $(a+b)^2 > 2c^2$.
Line 5. Since $a + b$ and c are positive, $a + b > \sqrt{2}\,c$.

Which line is the first that is not justified by the preceding argument?

A Line 3

B Line 1

C Line 5

D Line 2

E Line 4

WORKED SOLUTION

Lines 1-3 are valid. However, Line 3 gives an *upper* bound for $(a+b)^2$. Combining

$$(a+b)^2 \leq 2(a^2+b^2) \qquad \text{and} \qquad 2(a^2+b^2) > 2c^2$$

does not show that $(a+b)^2 > 2c^2$.

The claimed implication is false. For example, $a = 1.1$, $b = 0.1$, $c = 1$ gives $a^2 + b^2 = 1.22 > 1$, but $a + b = 1.2 < \sqrt{2}$.

Answer: E.

Paper 2 · Q11

Quadratic roots in an interval

Difficulty 5/5

Answer **C**

Question 11

Consider the quadratic

$$x^2 + ux + v.$$

Which set of conditions is necessary and sufficient for it to have two distinct roots, both lying in $(0,1)$?

A $u < 0,\ v > 0,\ u^2 > 4v$

B $-2 < u < 0,\ v < 0,\ u^2 > 4v,\ 1 + u + v > 0$

C $-2 < u < 0,\ v > 0,\ u^2 > 4v,\ 1 + u + v > 0$

D $-2 < u < 0,\ v > 0,\ u^2 > 4v,\ 1 + u + v < 0$

E $-2 < u < 0,\ v > 0,\ u^2 \geq 4v,\ 1 + u + v > 0$

WORKED SOLUTION

Let the roots be r, s. If $0 < r < s < 1$, then

$$u = -(r+s) \in (-2,0), \quad v = rs > 0, \quad u^2 - 4v > 0,$$

and

$$1 + u + v = (1-r)(1-s) > 0.$$

Conversely, the strict discriminant condition gives two distinct real roots. The signs of the sum and product make both roots positive. Their sum is below 2, so they cannot both exceed 1; and $f(1) > 0$ prevents exactly one root lying above 1. Hence both lie in $(0,1)$.

Answer: C.

Paper 2 · Q12

Functional equations & implication

Difficulty 3/5

Answer **D**

Question 12

A function $f : \mathbb{R} \to \mathbb{R}$ satisfies

$$f(x+y) = f(x) + f(y)$$

for all real x, y, and $f(x) \geq 0$ for every real x. Which statement must be true?

- **A** $f(x) = x$ for every real x
- **B** $f(x) = |x|$ for every real x
- **C** f is strictly increasing
- **D** $f(x) = 0$ for every real x
- **E** No such function exists

WORKED SOLUTION

Putting $x = y = 0$ gives $f(0) = 2f(0)$, so $f(0) = 0$.

For any real x,

$$0 = f(0) = f(x + (-x)) = f(x) + f(-x).$$

Hence $f(-x) = -f(x)$. But both $f(x)$ and $f(-x)$ are nonnegative, so the only possibility is $f(x) = 0$. This holds for every real x.

Answer: D.

Paper 2 · Q13 Implication & counterexample Difficulty 4/5 Answer **A**

Question 13

Let x be a nonzero real number. Consider

$$P: \ x \text{ is irrational}, \qquad Q: \ x + \frac{1}{x} \text{ is irrational}.$$

Which is correct?

A P is necessary but not sufficient for Q

B P is neither necessary nor sufficient for Q

C P is necessary and sufficient for Q

D Q is false for every x

E P is sufficient but not necessary for Q

WORKED SOLUTION

If x were rational and nonzero, then $1/x$ would also be rational, so $x + 1/x$ would be rational. Therefore

$$Q \implies P.$$

However, P does not imply Q. Take

$$x = \frac{3+\sqrt{5}}{2}.$$

This x is irrational and satisfies $x^2 - 3x + 1 = 0$. Dividing by $x \neq 0$ gives $x + 1/x = 3$, which is rational.

Thus P is necessary but not sufficient for Q.

Answer: A.

Paper 2 · Q14 Data sufficiency & prime exponents Difficulty 4/5 Answer **B**

Question 14

Let n be a positive integer. The objective is to determine whether $12 \mid n$. The following information is available:

$$\text{I.} \quad 18 \mid n^2, \qquad \text{II.} \quad 16 \mid n^3.$$

Which option is correct?

- **A** Even the two statements together are insufficient.
- **B** Statements I and II together are sufficient, but neither alone is sufficient.
- **C** Either statement alone is sufficient.
- **D** Statement II alone is sufficient.
- **E** Statement I alone is sufficient.

WORKED SOLUTION

From $18 = 2 \cdot 3^2 \mid n^2$, the exponent of 2 in n is at least 1, and the exponent of 3 is at least 1. Hence $6 \mid n$. This alone is insufficient: $n = 6$ satisfies I but is not divisible by 12.

From $16 = 2^4 \mid n^3$, if the exponent of 2 in n is r, then $3r \geq 4$, so $r \geq 2$. Hence $4 \mid n$. This alone is insufficient: $n = 4$ satisfies II but is not divisible by 12.

Together, $6 \mid n$ and $4 \mid n$, so $\text{lcm}(6, 4) = 12 \mid n$.

Answer: B.

Paper 2 · Q15 One-to-one functions Difficulty 3/5 Answer **C**

Question 15

Let $f : \mathbb{R} \to \mathbb{R}$. Consider

$$P: \ f \text{ is one-to-one (injective)},$$

$$Q: \ f \circ f \text{ is one-to-one (injective)}.$$

Which is correct?

- **A** P is necessary but not sufficient for Q
- **B** P is sufficient but not necessary for Q
- **C** P is necessary and sufficient for Q
- **D** P is neither necessary nor sufficient for Q
- **E** Q is impossible

WORKED SOLUTION

TMUA shortcut: The forward implication is automatic for compositions. For the converse, start from $f(a) = f(b)$ and apply f once more.

If f is one-to-one, then the composition of f with itself is one-to-one.

Conversely, suppose $f \circ f$ is one-to-one and $f(a) = f(b)$. Applying f gives

$$f(f(a)) = f(f(b)).$$

Since $f \circ f$ is one-to-one, $a = b$. Thus f itself is one-to-one.

Therefore P and Q are equivalent.

Answer: C.

Paper 2 · Q16 Symmetry & integration | Difficulty 4/5 | Answer **B**

Question 16

An integrable function f satisfies

$$f(-x) = 1 - f(x)$$

for every real x. Consider:

- I. $f(0) = \frac{1}{2}$.
- II. $\displaystyle\int_{-a}^{a} f(x)\,dx = a$ for every $a > 0$.
- III. $f(x+1) = f(x)$ for every real x.

Which statements must be true?

- **A** I only
- **B** I and II only
- **C** I, II and III
- **D** II only
- **E** III only

WORKED SOLUTION

Putting $x = 0$ gives $f(0) = 1 - f(0)$, so I is true.

Also, after pairing x and $-x$,

$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} \big(f(x) + f(-x)\big)\,dx = \int_{0}^{a} 1\,dx = a,$$

so II is true. The relation gives reflection symmetry, not periodicity, so III need not hold.

Answer: B.

Paper 2 · Q17 Triangle inequalities Difficulty 4/5 **Answer E**

Question 17

The side lengths of a triangle are

$$5, \qquad 5r, \qquad 5r^2.$$

What is the range of possible values of r?

A $0 < r < \dfrac{1+\sqrt{5}}{2}$

B $1 < r < \dfrac{1+\sqrt{5}}{2}$

C $\dfrac{\sqrt{5}-1}{2} \le r \le \dfrac{\sqrt{5}+1}{2}$

D $\dfrac{1-\sqrt{5}}{2} < r < \dfrac{1+\sqrt{5}}{2}$

E $\dfrac{\sqrt{5}-1}{2} < r < \dfrac{\sqrt{5}+1}{2}$

WORKED SOLUTION

TMUA shortcut: Scale out the common factor 5, then apply only the two triangle inequalities that can fail.

It is enough to consider side lengths $1, r, r^2$, with $r > 0$. The triangle inequalities are

$$1 + r > r^2, \qquad 1 + r^2 > r, \qquad r + r^2 > 1.$$

The middle inequality always holds because

$$r^2 - r + 1 = \left(r - \frac{1}{2}\right)^2 + \frac{3}{4} > 0.$$

The first and third inequalities give respectively

$$r < \frac{1+\sqrt{5}}{2}, \qquad r > \frac{\sqrt{5}-1}{2}.$$

The endpoints produce degenerate triangles, so both are excluded.

Answer: E.

Paper 2 · Q18 Number-theoretic proof Difficulty 4/5 **Answer D**

Question 18

Which expression is divisible by 30 for every integer n?

A $n^6 - n$

B $n^3 - n$

C $n^4 - n$

D $n^5 - n$

E $n^5 + n$

WORKED SOLUTION

TMUA shortcut: Factor first to obtain divisibility by 2 and 3; then check only the five possible residues modulo 5.

Factor

$$n^5 - n = n(n-1)(n+1)(n^2+1).$$

Among the three consecutive integers $n-1, n, n+1$, one is even and one is divisible by 3. Hence $n^5 - n$ is divisible by 6.

For divisibility by 5, every integer is congruent to $0, \pm 1$ or $\pm 2 \pmod 5$. In these five cases,

$$n^5 \equiv n \pmod 5,$$

because $0^5 = 0$, $(\pm 1)^5 = \pm 1$, and $(\pm 2)^5 \equiv \pm 2 \pmod 5$. Thus $5 \mid n^5 - n$.

Since $2, 3, 5$ are pairwise coprime,

$$30 \mid n^5 - n.$$

The other options fail, for example, at $n = 2$.

Answer: D.

Paper 2 · Q19 Symmetry & cancellation Difficulty 3/5 Answer **A**

Question 19

Let

$$f(x) = x^2(1-x)^2.$$

Find

$$f\left(\frac{1}{2019}\right) - f\left(\frac{2}{2019}\right) + f\left(\frac{3}{2019}\right) - \cdots + f\left(\frac{2017}{2019}\right) - f\left(\frac{2018}{2019}\right).$$

A 0

B $\dfrac{2018^2}{2019^4}$

C $-\dfrac{2018^2}{2019^4}$

D $\dfrac{1}{16}$

E $\dfrac{1}{2019^4}$

WORKED SOLUTION

TMUA shortcut: Do not expand the quartic or sum powers. Pair the k-th term with the $(2019-k)$-th term.

The function is symmetric about $x = \frac{1}{2}$:

$$f(1-x) = (1-x)^2x^2 = f(x).$$

Therefore

$$f\left(\frac{k}{2019}\right) = f\left(\frac{2019-k}{2019}\right).$$

The paired indices k and $2019-k$ have opposite signs in the alternating sum because their difference in parity is odd. Hence every term cancels with its partner.

$$\boxed{0}$$

Answer: A.

Paper 2 · Q20 Definitions & exhaustive checking Difficulty 3/5 **Answer B**

Question 20

A polygon is called **loud** if, for each occurrence of an interior angle A, there is a *distinct angle occurrence* B in the same polygon such that

$$B = \frac{A}{2} \qquad \text{or} \qquad B = 2A.$$

The occurrence chosen as B may have the same numerical measure as A, and one occurrence may serve as the required partner for more than one occurrence of A. A polygon that is not loud is called **quiet**.

Consider polygons with interior angles

$$\begin{aligned} \text{I.} &\quad 30^\circ, 60^\circ, 90^\circ, \\ \text{II.} &\quad 40^\circ, 80^\circ, 80^\circ, 160^\circ, \\ \text{III.} &\quad 60^\circ, 120^\circ, 120^\circ, 120^\circ, 120^\circ. \end{aligned}$$

Which polygons are quiet?

A II only

B I only

C II and III only

D I and II only

E III only

WORKED SOLUTION

TMUA shortcut: Check every distinct angle value, not merely whether one doubled pair exists somewhere.

For I, 30° and 60° can be paired, but 90° would require 45° or 180°, neither of which is present. So I is quiet.

For II, $40^\circ \leftrightarrow 80^\circ$ and $80^\circ \leftrightarrow 160^\circ$. Every angle occurrence has a suitable partner, so II is loud.

For III, $60^\circ \leftrightarrow 120^\circ$. The same 60° angle may witness the condition for each 120° occurrence, so III is loud.

Only I is quiet.

Answer: B.